
Presentation Transcript: Improving Wireless Charging Efficiency Using Photo-Sensing Multi-Transmitter Coil Arrays

[0:00 - 1:00] Introduction & Motivation "Good morning, judges and fellow researchers. Today, I am excited to present our project: Improving Wireless Charging Efficiency Using Photo-Sensing Multi-Transmitter Coil Arrays.

When we uncover challenges in our daily lives, it drives our imagination to find creative solutions. Our initial inspiration came from a common frustration: the energy inefficiency in current wireless charging technologies. Specifically, we identified two major problems. The first is 'Standby Drain'—chargers are always active, constantly wasting power even when not in use. The second is the 'Precision Gap'—if your phone isn't perfectly aligned, the misalignment leads to significant heat loss and inefficiency.

Our aim was to investigate the impact of various changing factors on receiver efficiency, construct a multi-coil wireless charging model, and ultimately improve upon commercially available solutions.

[1:00 - 2:30] Methodology & Pilot Experiments To build a solid foundation and ensure the feasibility of our goal, we started with pilot experiments. We tested three critical variables: frequency, distance, and offset. We didn't just want to prove an idea; we wanted to gather concrete results to guide our system design and logic.

[2:30 - 4:00] Data Interpretation: Distance, Frequency, and Offset *(Point to the 'Result' section, specifically the first two line graphs)* Let's look at the data. In our first test regarding vertical distance, we found that the receiver voltage exhibits exponential decay as the vertical distance increases. Similarly, when testing frequency, we observed that voltage decreases steadily as the operating frequency increases.

(Move to the 'Horizontal offset distance' graphs) However, the most crucial findings came from testing the horizontal offset distance. When using a single transmitting coil, as the offset increases, the voltage decays rapidly. Although we noticed a slight increase at a 5cm displacement, the receiving voltage unfortunately remains low.

But here is where our creative research plan comes in. We hypothesized that a multi-coil matrix design would solve this. When we tested an offset with *two* transmitting coils, we saw a different story. While there is still an overall downward trend, we observed significant local rebounds near the 2cm and 4.5cm marks. More importantly, between 1.5cm and 4.5cm, the receiver voltage is maintained between 4 to 7 Volts, before it continues to decrease after 5cm. This data perfectly validates that a multi-coil setup effectively expands the optimal charging area and eliminates blind spots.

[4:00 - 5:30] Smart System Design & Validation *(Point to the 'Arduino Code Design' bar chart)* Solving the precision gap was only half the battle; we still had to address the no-load consumption, or standby drain. To do this, we integrated photosensitive threshold settings using Arduino code design.

We collected values displayed by the photoresistor under different lighting conditions: indoor reference light, shadow, phone shielding, and no light. As you can see in the bar chart, when a phone shields the sensor, the light value drops. Based on this data, we set a precise threshold at a value of 50 to act as a trigger. By combining these photoresistor-controlled coil switches, the charger only activates the specific coils directly beneath the phone.

(Point to the 'Comparison' bar charts) To verify our integration, we ran an energy efficiency comparison. The results speak for themselves. In the comparison between our experimental group and a standard control group, our experimental group wins decisively in Receiver Voltage Performance. Furthermore, regarding no-load energy consumption, our experimental group wins again, proving effective power saving.

[5:30 - 7:00] Conclusion & SDGs In conclusion, by changing different fundamental factors, we successfully understood their impact on wireless charging. Using this experimental data, we successfully designed a highly efficient wireless charging model. We confirmed that combining photoresistor-controlled coil switches leads to effective power saving.

Finally, looking at the future potential and social value of our proposal, our design directly aligns with the United Nations Sustainable Development Goals, specifically SDGs 7 and 12. By enhancing charging efficiency and reducing battery wear, our research contributes significantly to e-waste reduction and sustainable energy consumption.

Thank you for your time and attention. I would now welcome any questions you may have."

Tips for your delivery:

- **Pacing:** Practice with a timer. If you speak too fast, pause longer when pointing at the graphs. The judges need a few seconds to process visual data.
- **Emphasis:** Emphasize words like *exponential decay*, *local rebounds*, and *no-load energy consumption*. These show your grasp of the scientific concepts.

這是一份為您量身打造的 15 題中英文問答速查表（Q&A Cheat Sheet）。這份清單涵蓋了研究動機、實驗變因、數據判讀以及永續發展目標（SDGs），能幫助您在 Tsukuba Science Edge 競賽的評審問答環節中對答如流。



Q&A 速查表：光感測多發射線圈陣列無線充電

1. 研究動機 (Motivation)

- **EN:** What inspired you to start this research on wireless charging?
- **CH:** 是什麼啟發您開始這項關於無線充電的研究？
- **Answer (EN):** We noticed energy inefficiency in wireless charging, specifically standby drain where it's always active and wasting power, and precision gaps where misalignment leads to heat loss.
- **Answer (CH):** 我們注意到無線充電的能源效率低落，特別是永遠保持開啟、浪費電力的「待機耗電」，以及對位不準確導致熱能流失的「對位落差」問題。

2. 核心問題 (Core Problem)

- **EN:** What exactly is the "Precision Gap" mentioned in your poster?
- **CH:** 您海報中提到的「對位落差 (Precision Gap)」具體是什麼意思？
- **Answer (EN):** It refers to the misalignment between the phone and the charger, which leads to heat loss and reduces charging efficiency.
- **Answer (CH):** 這是指手機與充電器之間沒有對準，這會導致熱能流失並降低充電效率。

3. 實驗方法 (Methodology)

- **EN:** What were the main variables you tested in your pilot experiments?
- **CH:** 在前導實驗中，您測試的主要變因有哪些？
- **Answer (EN):** We tested the impact of frequency, vertical distance, and horizontal offset on the receiver efficiency.
- **Answer (CH):** 我們測試了頻率、垂直距離以及水平偏移對接收端效率的影響。

4. 數據判讀：垂直距離 (Results: Vertical Distance)

- **EN:** How does vertical distance affect the receiver voltage?
- **CH:** 垂直距離如何影響接收電壓？

- **Answer (EN):** Based on our results, the receiver voltage exhibits exponential decay as the vertical distance increases.
- **Answer (CH):** 根據我們的結果，隨著垂直距離增加，接收電壓呈現指數衰減。

5. 數據判讀：頻率 (Results: Frequency)

- **EN:** What is the relationship between operating frequency and voltage?
- **CH:** 工作頻率與電壓之間有什麼關聯？
- **Answer (EN):** Our data shows that voltage decreases steadily as the operating frequency increases.
- **Answer (CH):** 我們的數據顯示，隨著工作頻率增加，電壓穩定下降。

6. 數據判讀：水平偏移 (Results: Horizontal Offset)

- **EN:** Why did you choose to use multiple transmitting coils instead of a single coil?
- **CH:** 為什麼您選擇使用多個發射線圈而不是單一線圈？
- **Answer (EN):** With a single coil, voltage decays rapidly. But with two transmitting coils, although there's a downward trend, local rebounds occur near 2cm and 4.5cm, and receiver voltage maintains between 4-7V between 1.5cm and 4.5cm.
- **Answer (CH):** 單一線圈的電壓衰減很快。但使用兩個發射線圈時，雖然整體呈下降趨勢，但在 2 公分與 4.5 公分附近會出現局部回升，且在 1.5 到 4.5 公分之間能維持 4-7 伏特的電壓。

7. 系統設計 (System Design)

- **EN:** What is the purpose of the photosensitive setting in your design?
- **CH:** 您的設計中加入光感測設定的目的是什麼？
- **Answer (EN):** We combined photoresistor-controlled coil switches to solve no-load consumption, confirming effective power saving.
- **Answer (CH):** 我們結合了光敏電阻控制的線圈開關來解決空載能耗，並證實能有效節省電力。

8. 程式設計 (Arduino Code Design)

- **EN:** How did you determine the threshold value of 50 for the photoresistor?
- **CH:** 您是如何決定光敏電阻的閾值為 50 的？
- **Answer (EN):** We evaluated values displayed by the photoresistor under different lighting conditions; the value drops to around 50 specifically during phone shielding.
- **Answer (CH):** 我們評估了光敏電阻在不同光線條件下的顯示數值；當手機遮蔽時，數值會特地降到 50 左右。

9. 效能比較 (Performance Comparison)

- **EN:** How does your experimental group compare to the control group?
- **CH:** 您的實驗組與對照組相比表現如何？
- **Answer (EN):** The experimental group wins in both receiver voltage performance and having lower no-load energy consumption.
- **Answer (CH):** 實驗組在接收電壓表現與較低的空載能耗方面雙雙勝出。

10. 聯合國永續發展目標 (SDGs Alignment)

- **EN:** How does your project align with the UN Sustainable Development Goals?
- **CH:** 您的專案如何與聯合國永續發展目標契合？
- **Answer (EN):** Our design aligns with SDGs 7 and 12 by enhancing charging efficiency and reducing battery wear, thereby contributing to e-waste reduction and sustainable energy consumption.
- **Answer (CH):** 我們的設計符合 SDG 7 和 12，藉由提升充電效率並減少電池耗損，從而對減少電子垃圾與永續能源消耗做出貢獻。

11. SDG 細節 (SDG Specifics)

- **EN:** Could you specify what SDG 7 and 12 stand for on your poster?
- **CH:** 您能具體說明海報上的 SDG 7 和 12 代表什麼嗎？

- **Answer (EN):** SDG 7 represents Affordable and Clean Energy, and SDG 12 represents Responsible Consumption and Production.
- **Answer (CH):** SDG 7 代表可負擔的潔淨能源，而 SDG 12 代表負責任的消費與生產。

12. 創新性 (Creativity & Originality)

- **EN:** What is the most novel aspect or matrix design of your research?
- **CH:** 您的研究中最具創新性的部分或矩陣設計是什麼？
- **Answer (EN):** Constructing a multi-coil wireless charging model integrated with photosensitive threshold settings to eliminate blind spots and save energy.
- **Answer (CH):** 建構多線圈無線充電模型，並整合光感測閾值設定，以消除死角並節省能源。

13. 商業應用 (Commercial Improvement)

- **EN:** How does your solution improve upon commercially available wireless charging solutions?
- **CH:** 您的解決方案如何改善市售的無線充電產品？
- **Answer (EN):** We improve them by reducing standby drain and precision gap issues through our multi-transmitter coil arrays and photo-sensing logic.
- **Answer (CH):** 我們透過光感測多發射線圈陣列及邏輯，減少待機耗電與對位落差的問題來改善市售產品。

14. 實驗意外發現 (Unexpected Results)

- **EN:** Were there any interesting data points during your horizontal offset tests?
- **CH:** 在水平偏移測試中，有任何有趣的數據點嗎？
- **Answer (EN):** Yes, with two transmitting coils, we observed local rebounds in voltage near 2cm and 4.5cm, which differs from a single coil's rapid decay.
- **Answer (CH):** 有的，使用兩個發射線圈時，我們觀察到電壓在 2 公分與 4.5 公分附近出現局部回升，這與單一線圈快速衰減的情況不同。

15. 結論 (Conclusion)

- **EN:** What is the ultimate takeaway from your research?
 - **CH:** 您的研究最終的結論是什麼？
 - **Answer (EN):** We successfully designed a set of wireless charging models using experimental data that confirms both effective power saving and enhanced charging efficiency.
 - **Answer (CH):** 我們成功利用實驗數據設計了一套無線充電模型，證實其能有效節省能源並提升充電效率。
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